PatchMix: Rethinking Patch Mixing for Small-Data Image Classification

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Abstract

We study patch-mixing augmentation for small-data image classification. On CIFAR-100 our method reaches 82.6 percent top-1 accuracy, beating the baseline by 1.9 points. This paper describes the full training setup for reproduction.

1. Architecture

Our model uses a ViT-style encoder with 12 layers and 8 attention heads. The hidden dimension (d_model) is 384 and the feed-forward dimension d_ff is 1536. We apply stochastic depth with rate 0.1.

2. Data and preprocessing

We train on CIFAR-100 using the standard train/test split. Images are resized to 32x32; we use RandAugment with magnitude 9. No test-time augmentation is used for the main results.

3. Training setup

The learning rate was set to 0.0008 with the AdamW optimizer, weight decay 0.05, and 10000 warmup steps under a cosine schedule. We train for 200 epochs with batch size of 512 examples. Dropout of 0.1 is applied to attention outputs. Random seeds 0, 1, 2 are averaged; we report the mean over 3 runs. Label smoothing of 0.1 is used with the cross-entropy loss. All models were trained on 4 NVIDIA A100 GPUs in bf16 precision.

4. Evaluation protocol

We report top-1 accuracy on the CIFAR-100 test split with a single central crop. The evaluation script version is v2. Checkpoint selection picks the best validation epoch.

Table 1: Main results

Model	Top-1 Acc (%)	Params (M)
Baseline ViT-S	80.7	22
PatchMix (ours)	82.6	22